

YUKI UENO

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Computer Science Ph.D. student at Arizona State University, specializing in Data Visualization and Human Computer Interaction, with five years of industry experience as a fullstack software engineer.

EDUCATION

2025.01 – present	Ph.D. in Computer Science, Arizona State University • Research Areas: Data Visualization and Human-Computer Interaction • Advisor: Prof. Chris Bryan	Tempe, AZ
2017.04 – 2019.03	M.Eng. in Electrical Engineering, Kyoto University • Research Areas: Data Visualization and Human-Computer Interaction • Advisors: Prof. Koji Koyamada and Prof. Hiroaki Natsukawa	Kyoto, Japan
2013.04 – 2017.03	B.Eng. in Electrical and Electronic Engineering, Kyoto University • Research Areas: Computer Vision, Human-Computer Interaction, and Human Sensing • Advisor: Prof. Yuichi Nakamura	Kyoto, Japan

PROFESSIONAL EXPERIENCES

2026.05 – 2026.08	Visual Data Scientist Intern, Eli Lilly and Company • Developed a mixed-initiative chart reconstruction system using MLLMs and an interactive verification interface • Benchmarked MLLMs for chart data and visual design extraction to guide system design	Indianapolis, IN
2025.01 – present	Graduate Research Associate, Arizona State University • Design and develop AI-assisted data visualization systems and integrate generative AI into interactive interfaces • Conduct user studies to evaluate system effectiveness	Tempe, AZ
2019.04 – 2024.12	Software Engineer, DENSO • Developed backend and frontend software for yuricargo • Developed software verification process standards for Arene OS in an international team • Developed a data analysis platform for software development management to support data-driven decision-making with various technologies, including Python, SQL, Tableau, and AWS • Developed a visualization system with ELK Stack to monitor application log data	Aichi, Japan

AWARDS AND HONORS

AWARDS

2025	Honorable Mention on IEEE PacificVis 2025 Visual Storytelling Contest. Yuki Ueno , Zhuojun Jiang, Bretho Danzy, Michael Kintscher, Nianwen Dan, Utkarsh Singh, and Chris Bryan. “The Story of Wagyu: Bringing Charm of Japan’s Pride to the World.” <i>IEEE PacificVis 2025 Visual Storytelling Contest</i> , 2025. demo
2025	Best Paper on The Visualization Society of Japan. Yuki Ueno , Hiroaki Natsukawa, and Koji Koyamada. “Do Boxes Affect Exploration Behavior and Performance in Group-in-a-box Layouts?” <i>Journal of Visualization</i> , Vol.28, pp. 449-462, 2025. https://doi.org/10.1007/s12650-024-01037-2

FELLOWSHIPS and SCHOLARSHIPS

2026-2027	Shida Scholarship , Kyodai Collaborative. Granted 40,000 USD
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TRAVEL GRANTS

2026	Conference Travel Award , Arizona State University. Support for conference registration and travel expenses
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PUBLICATIONS

JOURNAL

- 2026 Jinbin Huang, **Yuki Ueno**, Chen Chen, Aditi Mishra, Bum Chul Kwon, Zhicheng Liu, and Chris Bryan. “ASAP: Visual Analytics for Identifying and Analyzing Image Patterns in AI-generated Images” *Information Visualization*, 2026. <https://doi.org/10.1177/14738716261481077>
- 2025 **Yuki Ueno**, Hiroaki Natsukawa, and Koji Koyamada. “Do Boxes Affect Exploration Behavior and Performance in Group-in-a-box Layouts?” *Journal of Visualization*, Vol.28, pp. 449-462, 2025. [ **Best Paper**] <https://doi.org/10.1007/s12650-024-01037-2>
- 2019 **Yuki Ueno**, Hiroaki Natsukawa, Nozomi Aoyama, and Koji Koyamada. “Exploration behavior of group-in-a-box layouts.” *Journal of Visual Informatics* (also proc. *PacificVAST*), Vol. 3, No. 1, pp. 38-47, 2019. <https://doi.org/10.1016/j.visinf.2019.03.005>

CONFERENCE PAPERS

- 2026 **Yuki Ueno**, Bretho Danzy III, Zhuojun Jiang, and Chris Bryan. “VisCanvas: A Node-Based Interface for Exploratory Visualization Authoring with LLMs.” to be published in *IEEE TVCG / Proc. IEEE VIS 2026*, 2026. <https://doi.org/10.48550/arXiv.2607.21886>
- 2026 Zhuojun Jiang, **Yuki Ueno** and Chris Bryan. “Harnessing LLMs Without Surrendering Control: Delegation Boundaries in Visual Data Storytelling Authoring.” to be published in *Proc. IEEE VIS 2026 Short Paper*, 2026. <https://doi.org/10.48550/arXiv.2609.25700>
- 2019 Nozomi Aoyama, Yosuke Onoue, **Yuki Ueno**, Hiroaki Natsukawa, and Koji Koyamada. “User Evaluation of Group-in-a-Box Variants.” In *Proc. IEEE PacificVis*, pp. 127-136, 2019. <https://doi.org/10.1109/PacificVis.2019.00023>

WORKSHOP PAPERS and OTHER

- 2026 **Yuki Ueno** and Aditeya Pandey. “ChartRevive: Reconstructing Data Visualizations from Chart Images Using MLLM.” *GenAI, Agents, and the Future of VIS Workshop at IEEE VIS 2026*, 2026. <https://doi.org/10.48550/arXiv.2609.27146>
- 2025 Kentaro Takahira and **Yuki Ueno**. “VisAider: AI-Assisted Context-Aware Visualization Support for Data Presentations.” *MERCADO Workshop at IEEE VIS 2025*, 2025. <https://doi.org/10.48550/arXiv.2510.14247>
- 2025 **Yuki Ueno**, Zhuojun Jiang, Bretho Danzy, Michael Kintscher, Nianwen Dan, Utkarsh Singh, and Chris Bryan. “The Story of Wagyu: Bringing Charm of Japan’s Pride to the World.” *IEEE PacificVis 2025 Visual Storytelling Contest*, 2025. [ **Honorable Mention**] [demo](#)
- 2018 **Yuki Ueno**, Hiroaki Natsukawa, Nozomi Aoyama, and Koji Koyamada. “Task Performance Classification During Visualization Evaluations Based on Physiological Signals.” *VizAfrica 2018 Visualization Symposium*, 2018.
- 2018 Nozomi Aoyama, **Yuki Ueno**, and Koji Koyamada. “A Computational Evaluation of Eye-Tracking Measures in Group-in-a-Box Layouts.” *VizAfrica 2018 Visualization Symposium*, 2018.

RESEARCH EXPERIENCES

- 2025.10 – 2026.03 **VisCanvas** **Arizona State University**
Domain: Data Visualization, Visualization Authoring, Human-Computer Interaction
Advisors: Prof. Chris Bryan
- Developed an LLM-powered tool that support exploratory visualization authoring through a node-based interface
 - Utilized a fullstack architecture (Next.js, React, FastAPI) with LLM orchestration (LangGraph) deployed on AWS
 - Usability studies demonstrated more diverse data interaction while maintaining performance levels of linear interface

2025.06 – 2025.10	Augmented Algorithm Domain: Data Visualization, Human-Computer Interaction, Education Advisors: Prof. Chris Bryan - Developed an LLM-powered tool that converts static textbook pseudocode into interactive algorithm visualizations - Utilized a fullstack architecture (Next.js, React, FastAPI) with LLM orchestration (LangGraph) deployed on AWS - Usability studies and expert interviews demonstrated significant improvements in learning accessibility	Arizona State University
2025.01 – 2025.03	The Story of Wagyu Domain: Data Visualization, Storytelling Advisors: Prof. Chris Bryan - Developed a scroll-driven interactive visualization narrating the story of Japan's premium Wagyu beef - Utilized HTML, CSS, Bootstrap, D3.js, jQuery, and Scrollama to create an engaging storytelling experience - Awarded an <i>Honorable Mention</i> at the PacificVis 2025 Visual Data Storytelling Contest	Arizona State University
2017.04 – 2019.03	Evaluation of Group-in-a-Box Layout with Physiological Signals Domain: Data Visualization, Human-Computer Interaction, Cognitive Science Advisors: Prof. Koji Koyamada and Prof. Hiroaki Natsukawa - Investigated the behavioral patterns while interacting with graph drawings based on physiological signals - Compared performance of several graph-drawing methods based on user experiments	Kyoto University
2018.01 – 2018.02	Evaluation of Emotion with Physiological Signals Domain: Human-Machine Interface, Cognitive Science Advisor: Dr. Fumihiko Murase Investigated the relationship between subjects' emotions and physiological signals experienced while watching a video	DENSO
2016.04 – 2017.03	Modeling Machine Manipuration from Video Recording Domain: Computer Vision, Human-Computer Interaction, Human Sensing Advisor: Prof. Yuichi Nakamura Modeled manipulation of a sewing machine to automatically create a meaningful instructional manual from experience videos	Kyoto University

TEACHING EXPERIENCES

2026.9 – 2026.12	CSE478, 578 Data Visualization, Arizona State University (Graduate Course) Teaching Assistant	Tempe, AZ
2018.04 – 2018.07	ILAS Seminar in Visualization Lab, Kyoto University (The ILAS Seminar is one of the Liberal Arts and Sciences Courses mainly for new undergraduate freshmen, taught in a small-group seminar format conducted by professors. The seminar topics included methods for data analysis of multidimensional data.) Teaching Assistant	Kyoto, Japan
2017.10 – 2018.01	Visualized Simulation Technology, Kyoto University (Graduate Course) Teaching Assistant	Kyoto, Japan
2017.04 – 2017.07	Spacio-temporal Data Analysis for Multimedia, Kyoto University (Graduate Course) Teaching Assistant	Kyoto, Japan

ACADEMIC SERVICE

2026	Student Volunteer, IEEE VIS 2026
2026	External Reviewer, IEEE VIS 2026
2026	External Reviewer, ACM CHI 2026
2025	External Reviewer, IEEE PacificVis 2026
2025	External Reviewer, IEEE VIS 2025
2018	Student Volunteer, IEEE PacificVis 2018

CERTIFICATIONS

2021	AWS Certified SysOps Administrator – Associate
2021	AWS Certified Solutions Architect – Associate
2020	AWS Certified Cloud Practitioner
2020	Applied Information Technology Engineer (Certification in Japan)

SKILLS

Programming	Python, JavaScript, TypeScript, Java, C, SQL, Bash
Frontend	HTML, CSS, Tailwind CSS, React, Next.js, Node.js, D3.js
Tools	FastAPI, LangGraph, PostgreSQL, AWS, Docker, Serverless Framework, Elasticsearch, Tableau, GitHub